

MNYMO Competition 2024

Estimathon Solutions

MINNESOTA YOUTH MATH OUTREACH

August 31, 2024

Problem 1. Estimate the population density of penguins in Antarctica, in penguins per square mile.

Answer: 8. The penguin population is estimated at 44,000,000 by the World Population Review. Moreover, Antarctica is approximately 5,500,000 square miles in area. $44,000,000 / 5,500,000 = 8$.

Problem 2. Estimate the geometric mean of Spotify streams Taylor Swift has on her most popular album and the least popular album as of August 21, 2024.

Answer: 3,995,541,487. Taylor Swift's 2019 album "Lover" has the most streams on Spotify with 11,034,616,762. Her 2006 album "Taylor Swift" has the least streams on Spotify with 1,446,751,810. The geometric mean is approximately $\sqrt{1446751810 \cdot 11034616762} = 3,995,541,487$.

Problem 3. Estimate the sum of the elevation of the 30 mountains in the United States with the greatest topographic elevation in feet.

Answer: 451,611. From the tallest to shortest, the 30 tallest peaks in North America have the height of (in feet): $20310 + 18009 + 17400 + 16550 + 16390 + 16237 + 15325 + 14951 + 14831 + 14573 + 14505 + 14500 + 14470 + 14440 + 14428 + 14421 + 14417 + 14379 + 14351 + 14343 + 14321 + 14300 + 14293 + 14279 + 14278 + 14276 + 14271 + 14259 + 14252 + 14252 = 451,611$.

Problem 4. Estimate the number of times the phrase "slim shady" occurs in the lyrics of "The Real Slim Shady" by Eminem.

Answer: 29.

Problem 5. Estimate the area of the quadrilateral formed by connecting the current home stadiums of the last 4 NFL championship-winning teams in square miles (disregarding repeat championships).

Answer: 710,009. The last four champions are the Kansas City Chiefs (SB LVIII), the Los Angeles Rams (SB LVI), the Tampa Bay Buccaneers (SB LV), and the New England Patriots (SB LIII). Their home stadiums are the Arrowhead Stadium (Kansas City, Missouri), the SoFi Stadium (Inglewood, California), the Raymond James Stadium (Tampa, Florida), and the Gillette Stadium (Foxborough, Massachusetts), respectively. Connecting these four stadiums, the quadrilateral area is approximately 710,009 square miles.

Problem 6. Estimate the surface area of the Eiffel Tower in square inches. (Only include exterior surface area; if you were to repaint the tower, how much paint would you use?)

Answer: 387,500,775. A rough estimate of the Eiffel Tower's surface area is 250,000 square meters. 1 square meter is roughly 1550.0031 square inches, so multiplying we get about 387,500,775 square inches.

Problem 7. Estimate the number of prime numbers less than 1,000,000 (1 million).

Answer: 78,498.

Problem 8. Estimate the length of the longest border any country shares with China, in miles.

Answer: 2,880. Mongolia shares the longest border with China with a length of 2,880 miles.

Problem 9. Estimate the number of silver medals the United States has won in both summer and winter Olympic events since 2000.

Answer: 313. Ordering from Sydney 2000 to Paris 2024: $24 + 13 + 39 + 9 + 39 + 15 + 26 + 9 + 37 + 8 + 41 + 9 + 44 = 313$.

Problem 10. Estimate the number of nickels minted in 2023.

Answer: 1,427,500,000.

Problem 11. Estimate the number of graphing calculators Texas Instruments has sold since 1990.

Answer: 90,000,000.

Problem 12. Estimate the number of views of Cocomelon's most viewed video on YouTube as of August 21, 2024.

Answer: 6,823,524,311. "Bath Song @CoComelon Nursery Rhymes & Kids Songs" has garnered nearly 7 billion views.

Problem 13. Estimate the value of the quotient when the mean of the answers to all previous problems is divided by the median.

Answer: 1825.8. The mean is 1,060,442,492 and the median is 580,810. $1,060,442,492 / 580,810 = 1825.79930098$.